

The Millennium

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Abstract

What we knew at the turn of the century. What we still do not know.

In the year 2000 we knew the following.

We knew the Riemann Hypothesis was probably true. We had verified it for the first ten billion zeros. We had proved it was equivalent to dozens of things. We had built entire fields of mathematics on the assumption it was true. We had not proved it.

We knew Yang–Mills theory described three of the four fundamental forces. We had calculated with it to eleven decimal places. We had won Nobel Prizes for work that assumed it was a well-defined quantum theory. We had not proved it was a well-defined quantum theory.

We knew P was probably not equal to NP . We had built the entire edifice of cryptography on this assumption. The security of every encrypted message on earth depended on it. We had not proved it.

A millennium is a unit of time chosen by humans for humans. Mathematics does not respect it. The Riemann Hypothesis was not harder in 2000 than in 1900. The walls did not get higher because the calendar turned.

What the millennium gave us: a list. Seven problems, named and priced, the difficulty acknowledged formally, the reward offered publicly.

A list is not a solution. A list is a way of looking at what you do not know and agreeing that not knowing is worth something. The Clay Institute said: these are the right problems. We do not know the answers. We are willing to pay to find them.

That is a civilizational act.

Twenty-five years later, one of the seven has been solved. Perelman proved Poincaré. The other six stand.

The millennium continues. The problems do not care.